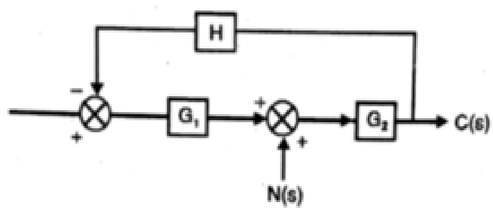
CONTROL SYSTEMS – UE24EE241B

Sample MCQs

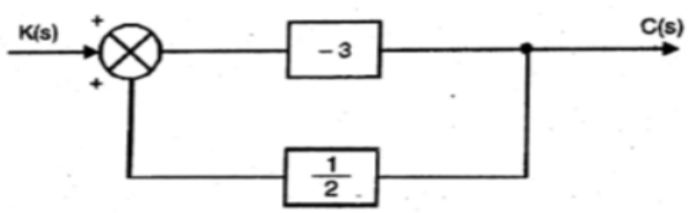
BLOCK DIAGRAMS

1. The closed loop system shown in figure is subjected to a disturbance N(s). The transfer function C(s)/N(s) is given by



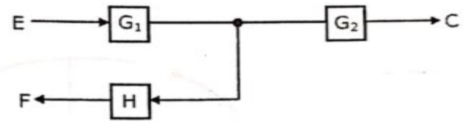
- $\frac{C(s)}{N(s)} = \frac{G_1}{1+G_2G_1H}$
- $\frac{C(s)}{N(s)} = \frac{G_2}{1+G_2G_1H}$
- $\frac{C(s)}{N(s)} = \frac{G_2}{1+G_2H}$
- $\frac{C(s)}{N(s)} = \frac{G_2}{1+G_2G_1}$

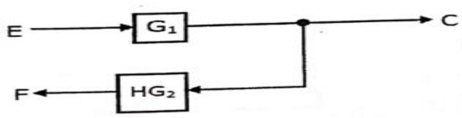
2. The closed loop gain of the system shown in the given figure is

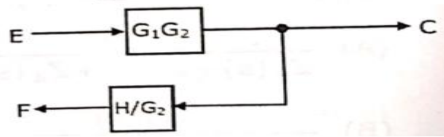


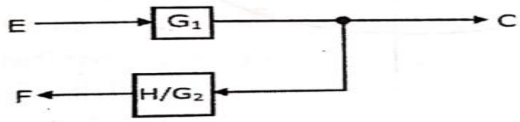
- 6/5
- -9/5
- 9/5
- -6/5

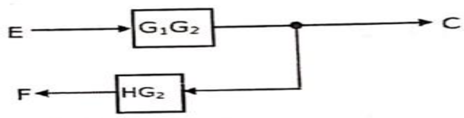
3. Identify the equivalent of the block diagram shown in figure:





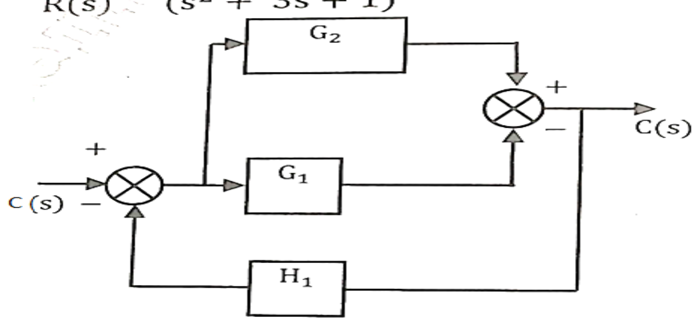






4. Consider the block diagram shown below. It's given that system has unity feedback ($H_1 = 1$). Then which of the following is possible G_1 and G_2 if

$$\frac{C(s)}{R(s)} = \frac{(s-1)}{(s^2 + 3s + 1)}$$



(A) $\frac{s}{(s^2 + 2s + 2)} \& \frac{1}{(s^2 + 2s + 2)}$

(B) $\frac{1}{(s^2 + 2s + 2)} \& \frac{s}{(s^2 + 2s + 2)}$

(C) $\frac{s}{(s^2 + 3s + 1)} \& \frac{1}{(s^2 + 3s + 1)}$

(D) $\frac{1}{(s^2 + 3s + 1)} \& \frac{s}{(s^2 + 3s + 1)}$

Date: 27-01-26 **Dr. Anuradha M**
(Anchor)